SkyPulse AI

Every single minute of delay costs $187.

A unified intelligence operational platform built entirely on Snowflake's AI Data Cloud, transforming operational losses into a $47.5M value opportunity.

Snowflake AI Innovation Day | London 2026

Four Challenges, One Platform

The Operational Friction Root Cause

These distinct structural issues are deeply interconnected. Data silos across Flight Operations, Customer Experience, and Engineering lead to cascading reactive decisions rather than predictive strategy.

- Flight delays compound into customer loyalty churn
- Weak revenue overbooking forecasts lose margins
- Reactive component maintenance creates unplanned AOGs

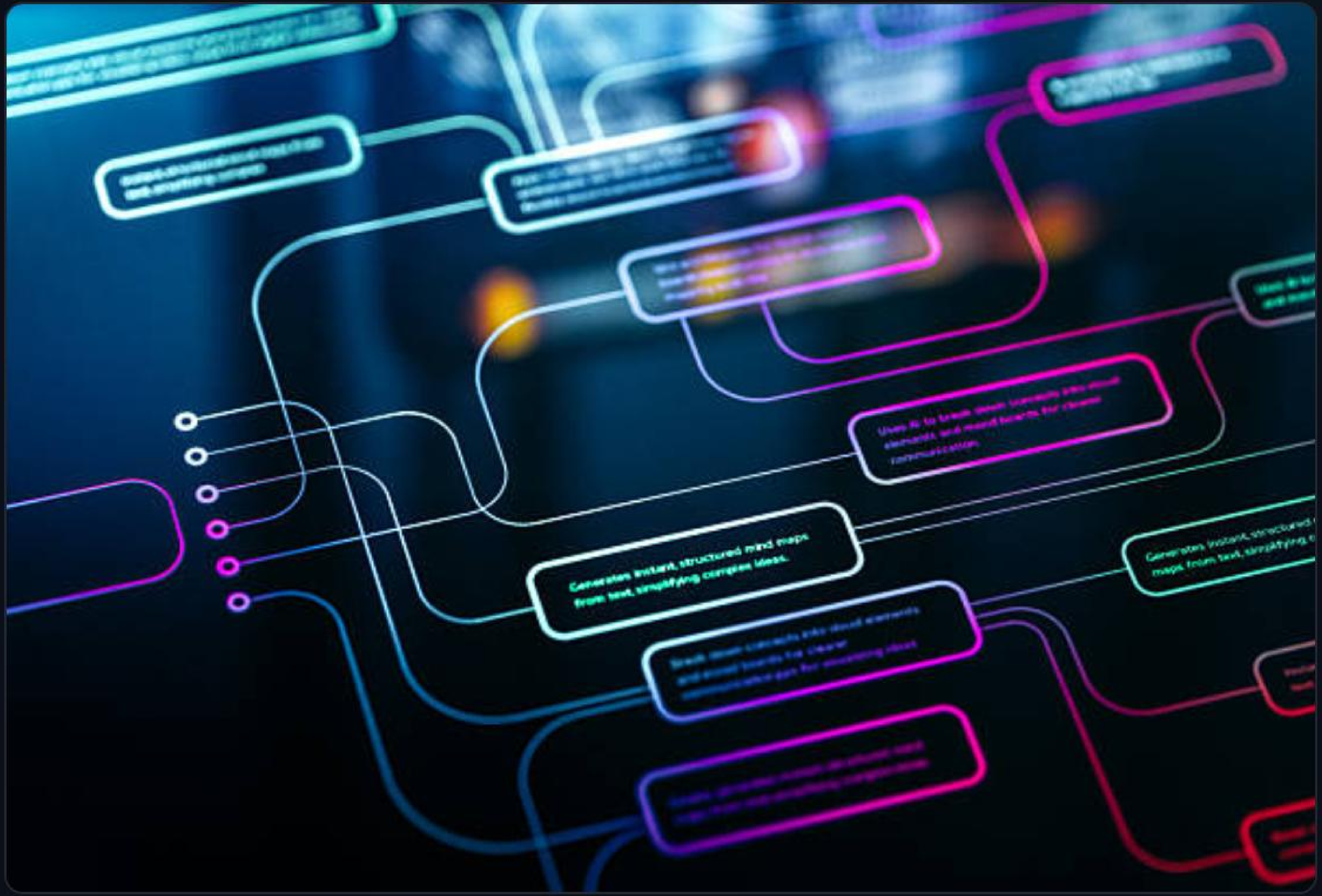
Airline Challenge Dimension	Annual Value Leakage
Flight Delays & Compensation (EU261)	$95,000,000
Customer Churn (High-Value Members)	$52,000,000
Overbooking & Revenue Leakage	$28,000,000
Reactive Unscheduled Maintenance	$12,000,000
Cumulative Annual Cost Impact	**$187,000,000**

Unified Intelligent Architecture

Zero External ML Infrastructure Required

Built natively on Snowflake's AI Data Cloud using **Medallion Architecture** pipelines. This provides an end-to-end continuous business logic engine without data transport latency.

- **Bronze Layer:** Streaming raw feedback, telemetry, and bookings.
- **Silver Layer:** Structured star schema with CDC state engines.
- **Gold Layer:** Low-latency business views optimized for AI.

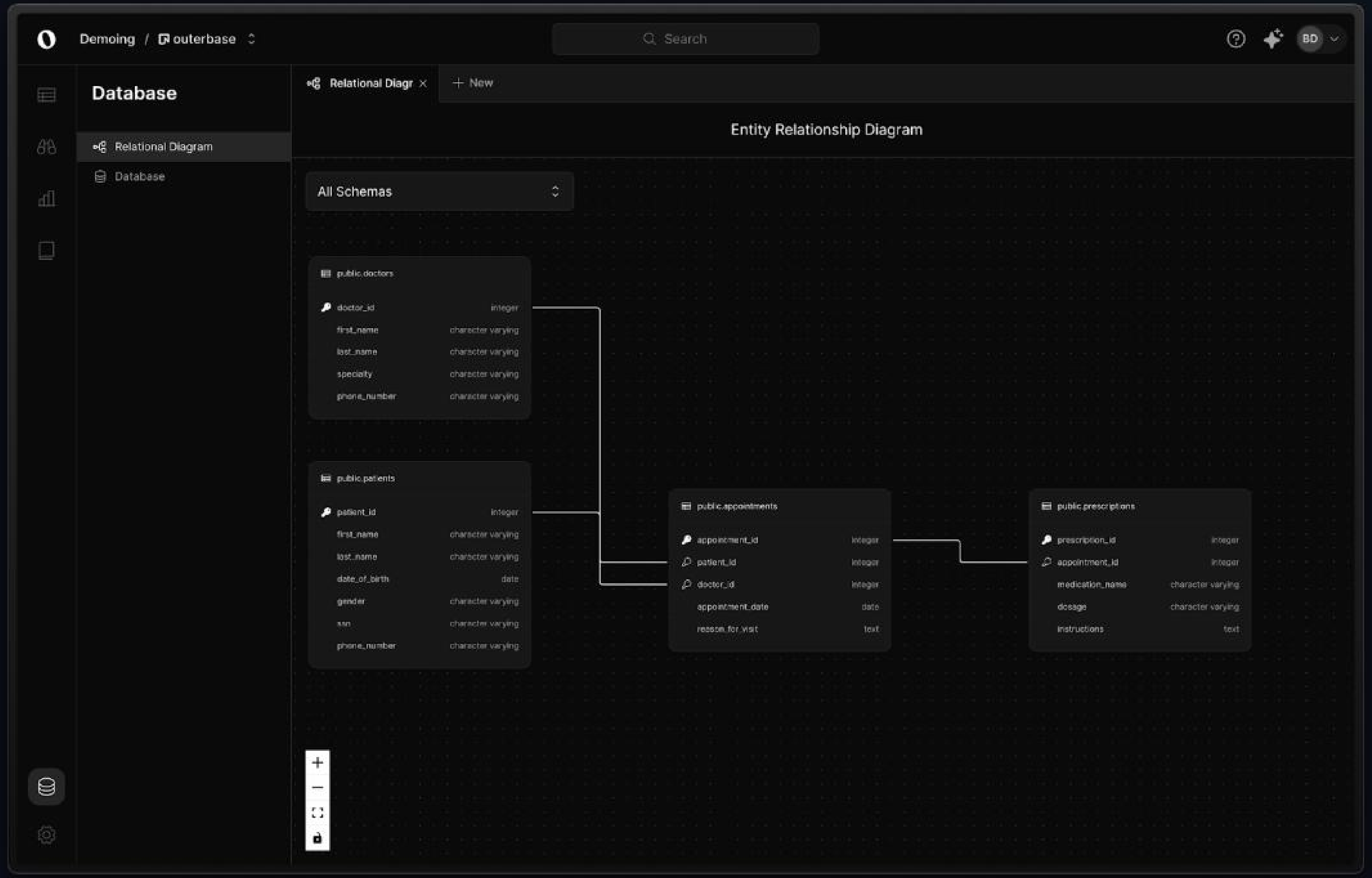


Enterprise-Grade Star Schema

Optimized Dimensional Database Model

Our data platform coordinates transactional, operational, and customer records dynamically to yield rapid analytics capabilities.

- **Core Dimensions:** Passengers (SCD Type 2), Airports, Aircraft types, and Routes.
- **Fact Transactions:** Live Flights, Bookings, Disruptions, and Service Feedback.
- **Built-in Governance:** Tagged PII columns with dynamic row-level masking policies.



Live Demo: Cortex AI Sentiment

1. Ingest & Score

Extract sentiment directly inside SQL pipelines without external API routing.

`CORTEX.SENTIMENT(feedback)`

2. Extract Summary

Distill long multi-paragraph operational complaints into clear, actionable bullet points.

`CORTEX.SUMMARIZE(complaint)`

3. Service Recovery

Generate dynamic, personalized apology responses complete with appropriate vouchers.

`CORTEX.COMPLETE(prompt)`

Live Demo: Predictive Cortex ML

Delay Forecasts

Forecasts route congestion in advance.

`LHR-JFK Tuesday: 28-min delay predicted. Rebook now.`

Churn Classification

Detects loyalty attrition thresholds.

`Diamond Member Churn Probability: 82% (Send alert)`

Anomaly Detection

Flags early turbine thermal signatures.

`AOG risk identified on G-SPDA (15% fuel burn anomaly)`

Real-time Operations Pipeline

Dynamic Tables & CDC

Our pipelines leverage **Dynamic Tables** to automatically refresh core flight statistics every single minute, maintaining up-to-date data structures.

- Low-latency data updates trigger operational status changes instantly.
- Automatic system notifications are dispatched when OTP drops below 75%.

Automated Streams & Tasks

Data changes on incoming tickets trigger downstream Cortex processes within 10 minutes, generating live gate assignment optimizations.

- UDF functions perform micro-batch model inferences natively.
- Fully structured to scale dynamically with workload warehouse isolation.

12 Native Snowflake Features

Feature Element	Specific Aviation Use Case	Feature Element	Specific Aviation Use Case
Cortex LLM	Generates loyalty member sentiment & summaries	**Dynamic Tables**	Near real-time materialized views
Cortex Forecast	Aviation route delay predictions	**Snowpark Python**	Data engineering & modeling UDFs
Cortex Anomaly	Unplanned aircraft thermal anomaly detection	**Streams & Tasks**	Automated operational pipeline execution
Cortex Classify	Loyalty membership attrition prediction	**Dynamic Masking**	Strict GDPR compliance and passenger PII protection
System Alerts	Automated triggers for on-time arrivals drops	**Data Sharing**	Operational data handoffs with regional airports
Multi-Cluster	Workload isolation (Operations vs. Analytics)	**Time Travel**	Compliance audits and flight histories retrieval

Business Impact: 22.6x ROI

$47.5M

Net Year 1 Program Value

Calculated on total investment of $2.1M (Infrastructure & Resources)

Operations Strategy Value Stream	Projected Savings
Delay Forecasting & Proactive Passenger Rebooking	$28,500,000
High-Value Loyalty Attrition & Churn Prevention	$6,200,000
Auto-Cortex Sentiment Service Recovery	$8,000,000
Telemetry Anomalies Predictive Aircraft Maintenance	$4,800,000
Annual Net Optimization Value	**$47,500,000**

Implementation Roadmap

Weeks 1-4

Ingestion & Integration

Establishing real-time ACARS telemetry feed, Amadeus booking interfaces, and airport CDC streams inside Snowflake Bronze.

Weeks 5-8

Cortex Model Tuning

Optimizing prediction latency, validating forecasting accuracy metrics, and tuning apology prompt templates natively inside SQL.

Weeks 9-12

Deployment & Value

Launching operational Streamlit portals for gate managers, flight planners, and engineering teams to achieve rapid operational value.

Questions & Answers

Because every passenger deserves a flight that runs on intelligence, not just fuel.

```
demo-workspace: SKYPULSE_ML_WH
repository: github.com/skypulse/ai-innovation-day
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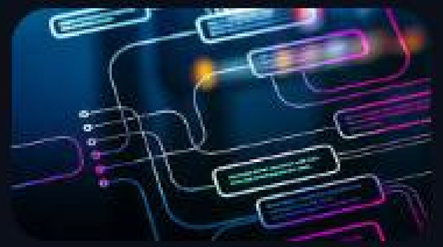
Image Sources

https://png.pngtree.com/png-vector/20250214/ourlarge/pngtree-futuristic-blue-commercial-airplane-with-modern-design-png-image_15454309.png

Source: pngtree.com



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Source: neon.com